



Acid Rain Detectives

My name _____

Class _____

Date _____



How can we detect acid, and what does acid rain do to stone?

Smoke from factories and cars can make rain a little bit acidic. Acid rain can harm plants and wear away stone. Let's become acid detectives!



My kit tick when you have it

- ☐ A few red cabbage leaves
- ☐ Hot water (an adult pours it)
- ☐ A bowl, a sieve and a jug
- ☐ Three clear cups
- ☐ Water, vinegar and baking soda water
- ☐ Two limestone chips or eggshell pieces
- ☐ Two clear cups
- ☐ Water and vinegar



Stay safe

- Only an adult uses the knife and hot water.
- Wear safety glasses.
- Never drink or taste anything in science.
- Wash your hands when you finish.
- Tell an adult if anything splashes in your eyes.

START HERE: BUILD



Make your colour tester

1 · DO IT: TICK EACH STEP

- 1 ☐ An adult chops the cabbage into small pieces.
- 2 ☐ An adult pours hot water over it.
- 3 ☐ Wait until the water is cool and purple.
- 4 ☐ Pour it through a sieve into a jug.

2 · LOOK & RECORD: CIRCLE WHAT YOU SEE

	What colour did it turn?
My colour tester	Pink or red Purple Blue or green



Draw your tester

WORDS TO USE

cabbage purple colour tester sieve cool

TEST 1



Colour detectives

? Which liquid will turn the tester pink?

1 · MY GUESS: CIRCLE ONE

☐ Plain water ☐ Vinegar ☐ Baking soda water

How sure am I? ☐ ☐ ☐

2 · DO IT: TICK EACH STEP

- ☐ Put a little of each liquid in its own cup.
- ☐ Label the cups so you remember which is which.
- ☐ Add a spoon of colour tester to each cup.
- ☐ Look at each colour. Tap what you see.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	What colour did it turn?
☐ Water	☐ Pink or red ☐ Purple ☐ Blue or green
☐ Vinegar	☐ Pink or red ☐ Purple ☐ Blue or green
☐ Baking soda water	☐ Pink or red ☐ Purple ☐ Blue or green

4 · WHAT HAPPENED? CIRCLE ONE

☒ My guess was right ☐ Partly right ☐ Something different happened ☐ Not sure yet

I found out that _____

because _____

WORDS TO USE

acid pink purple blue green vinegar soda indicator

TEST 2 Acid on stone

? What happens to stone in vinegar?

1 · MY GUESS: CIRCLE ONE

☐ Water ☐ Vinegar ☐ Both the same

How sure am I? ☐ ☐ ☐

2 · DO IT: TICK EACH STEP

- ☐ Pour water into one cup.
- ☐ Pour vinegar into the other cup.
- ☐ Drop one stone into each cup.
- ☐ Watch closely for 5 minutes.
- ☐ Tap how many bubbles you see in each cup.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	How many bubbles?
 Stone in water	☐ None ☐ A few ☐ Lots
 Stone in vinegar	☐ None ☐ A few ☐ Lots

 Draw what you see: Stone in water

 Draw what you see: Stone in vinegar

4 • WHAT HAPPENED? CIRCLE ONE

☒ My guess was right
 ☐ Partly right
 ☐ Something different happened
 ☐ Not sure yet

I found out that _____

because _____

WORDS TO USE

fizz bubbles stone acid vinegar dissolve statue

 **Art corner: Magic colour painting**

Paint a picture with cabbage water. Dab on vinegar for pink and baking soda water for green. Stick or draw it here.

 **Maths corner: Colour table**

Count the bubbles you see in 10 seconds in each cup: water ___, vinegar ___.

 **New words**

Indicator: Something that changes colour to tell us about a liquid. **Acid:** A sour kind of liquid, like vinegar or lemon juice. **Base:** The opposite of an acid, like baking soda.

Dissolve: When something breaks down and mixes into a liquid.

★ **What I can do now** colour the stars

I can use a colour tester to find an acid.



I can sort liquids by the colour they make.



I can explain how acid rain damages stone.

